

Question Paper: Ai

****Artificial Intelligence (AI) Question Paper****

****Medium Difficulty****

****Instructions:****

- * The question paper consists of three sections: Multiple Choice Questions (MCQs), Short Questions, and Long/Descriptive Questions.
- * Attempt all questions in each section.
- * Use a separate sheet for each section.

****Section A: Multiple Choice Questions (30 marks)****

1. Which of the following is a type of machine learning algorithm?

- A) Rule-based system
- B) Decision tree
- C) Support vector machine
- D) All of the above

Correct answer: D) All of the above

2. What is the primary function of a neural network in AI?

- A) To process natural language
- B) To recognize patterns in data

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C) To make decisions based on rules

D) To optimize system performance

Correct answer: B) To recognize patterns in data

3. Which AI technique is used to analyze and understand human language?

A) Natural Language Processing (NLP)

B) Computer Vision

C) Robotics

D) Expert Systems

Correct answer: A) Natural Language Processing (NLP)

4. What is the term for a computer system that can perform tasks that typically require human intelligence?

A) Machine learning

B) Artificial intelligence

C) Deep learning

D) Natural language processing

Correct answer: B) Artificial intelligence

5. Which of the following is an application of AI in the healthcare industry?

A) Virtual assistants

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- B) Medical diagnosis
- C) Personalized medicine
- D) All of the above

Correct answer: D) All of the above

6. What is the name of the AI system that defeated a human world champion in chess in 1997?

- A) Deep Blue
- B) AlphaGo
- C) Watson
- D) Siri

Correct answer: A) Deep Blue

7. Which AI algorithm is used for unsupervised learning?

- A) K-means clustering
- B) Linear regression
- C) Decision tree
- D) Support vector machine

Correct answer: A) K-means clustering

8. What is the term for a computer system that can learn from experience and improve its performance over time?

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- A) Machine learning
- B) Artificial intelligence
- C) Deep learning
- D) Neural network

Correct answer: A) Machine learning

9. Which of the following is a challenge in developing AI systems?

- A) Data quality
- B) Algorithm complexity
- C) Ethics and bias
- D) All of the above

Correct answer: D) All of the above

10. What is the name of the AI system that can simulate human-like conversations?

- A) Chatbot
- B) Virtual assistant
- C) Natural language processing
- D) Dialogue system

Correct answer: A) Chatbot

****Section B: Short Questions (30 marks)****

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1. What is the difference between narrow or weak AI and general or strong AI? (3 marks)
2. Explain the concept of supervised learning in machine learning. (4 marks)
3. Describe the application of AI in the field of robotics. (4 marks)
4. What is the role of bias in AI decision-making? (3 marks)
5. Explain the concept of deep learning and its applications. (5 marks)

****Section C: Long/Descriptive Questions (30 marks)****

1. Describe the architecture of a neural network and explain its components. How does it learn from data? (8 marks)
2. Discuss the applications and limitations of natural language processing (NLP) in AI. Provide examples of NLP in real-world scenarios. (9 marks)
3. Explain the concept of machine learning and its types (supervised, unsupervised, and reinforcement learning). Provide examples of each type and their applications. (10 marks)

****Answer Key:****

Section A: MCQs

1. D) All of the above
2. B) To recognize patterns in data
3. A) Natural Language Processing (NLP)
4. B) Artificial intelligence

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- 5. D) All of the above
- 6. A) Deep Blue
- 7. A) K-means clustering
- 8. A) Machine learning
- 9. D) All of the above
- 10. A) Chatbot

Section B: Short Questions

- 1. Narrow or weak AI refers to AI systems that are designed to perform a specific task, whereas general or strong AI refers to AI systems that can perform any intellectual task that a human can.
- 2. Supervised learning is a type of machine learning where the AI system is trained on labeled data to learn the relationship between input and output.
- 3. AI is used in robotics to enable robots to perform tasks such as object recognition, navigation, and manipulation.
- 4. Bias in AI decision-making refers to the unfair or discriminatory outcomes that can occur when AI systems are trained on biased data or designed with biased algorithms.
- 5. Deep learning is a type of machine learning that uses neural networks with multiple layers to learn complex patterns in data. Its applications include image and speech recognition, natural language processing, and game playing.

Section C: Long/Descriptive Questions

- 1. A neural network consists of an input layer, hidden layers, and an output layer. It learns from data

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through a process called backpropagation, where the error between predicted and actual outputs is minimized.

2. NLP has applications in language translation, sentiment analysis, and text summarization. However, it is limited by the quality of the training data and the complexity of human language.

3. Machine learning is a type of AI that enables systems to learn from data without being explicitly programmed. Supervised learning involves training on labeled data, unsupervised learning involves discovering patterns in unlabeled data, and reinforcement learning involves learning from rewards or penalties. Examples of each type include image classification, clustering, and game playing.